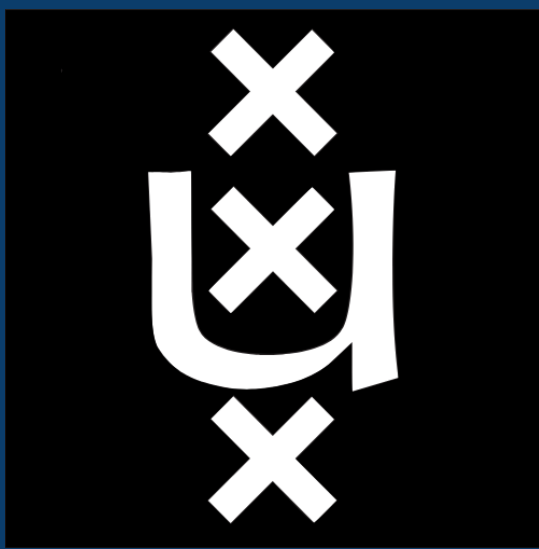


A comprehensive framework for Bayesian meta-analysis

Robust Bayesian meta-analysis: RoBMA R package & JASP Meta-Analysis module

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Bayesian meta-analysis

Few studies

Informed priors

Single model inference

Model averaging

No sampling plan

Bayes factors

Null findings

Evidence for H0

Publication bias

Competing adjustments

Bayesian model averaging

Effect

Heterogeneity

Moderators

Publication bias

Bayesian
model averaging
product-space via JAGS

Model-averaged
estimates

Inclusion
Bayes factors

Prior distributions

- Default (UISD-based)
auto-scaled to measure
- Empirical / informed
Cochrane medicine, psychology
- Custom
Flexible constructors for known distributions
- Publication-bias
Selection models, PET, PEESE

Feature overview

Model fitting

```
brma() · brma.glmm()
BMA() · RoBMA()
bselmodel() · bPET() · bPEESE()
```

Priors

```
prior() · prior_factor()
prior_informed()
prior_weightfunction()
```

Model comparison

```
loo() · waic()
loo_compare() · loo_weights()
bayes_factor() · post_prob()
```

Estimates & prediction

```
pooled_effect() · marginal_means()
predict() · fitted()
ranef() · blup()
```

Plots

```
plot() · plot_prior()
funnel() · forest() · regplot()
zplot() · plot_weightfunction()
```

Diagnostics

```
residuals() · rstudent()
cooks.distance() · influence()
vif() · check_loo()
```

MCMC diagnostics

```
summary() — Rhat, ESS
trace · density · autocorr.
```

Reporting & extraction

```
interpret() · summary_models()
as_draws_*(()) · coef()
```

Example



Havránková (2025); Beauty & professional success · 1,159 effect sizes / 67 studies

```
library(RoBMA)
data("Havrankova2025")

fit <- RoBMA(
  yi = y, sei = se, ni = N,
  mods = ~ facing_customer, cluster = study_id,
  prior_effect = prior("normal", list(mean = 0, sd = 12.9)),
  prior_heterogeneity = prior("normal", list(mean = 0, sd = 6.45), truncation = list(0, Inf)),
  prior_mods = list(facing_customer = prior_factor("mnormal", list(mean = 0, sd = 6.45), contrast = "meandif")),
  data = Havrankova2025, measure = "GEN", seed = 1,
  parallel = TRUE, adapt = 10000, burnin = 10000, sample = 20000)
```

```
> summary(fit)

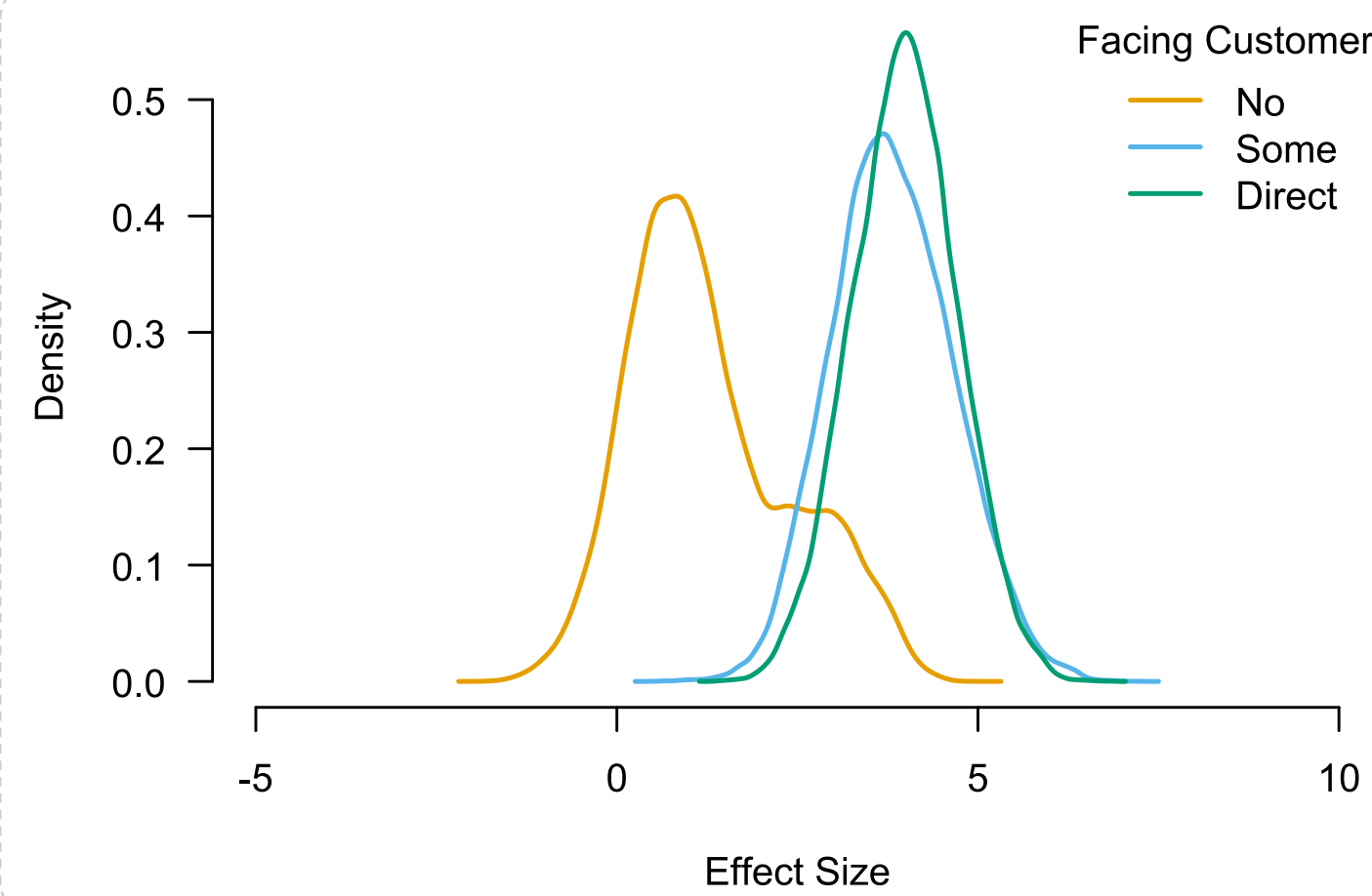
Robust Bayesian Model-Averaged Multilevel Mixed-Effect Model
(k = 1159, clusters = 67)

Component Inclusion      Prior prob. Post. prob. Inclusion BF
Effect                  0.500      1.000 >79999.000
Heterogeneity           0.500      1.000 >79999.000
Publication Bias         0.500      1.000 >79999.000

Meta-Regression Inclusion
facing_customer          Prior prob. Post. prob. Inclusion BF
                        0.500      0.853      5.795

Common Estimates
...
```

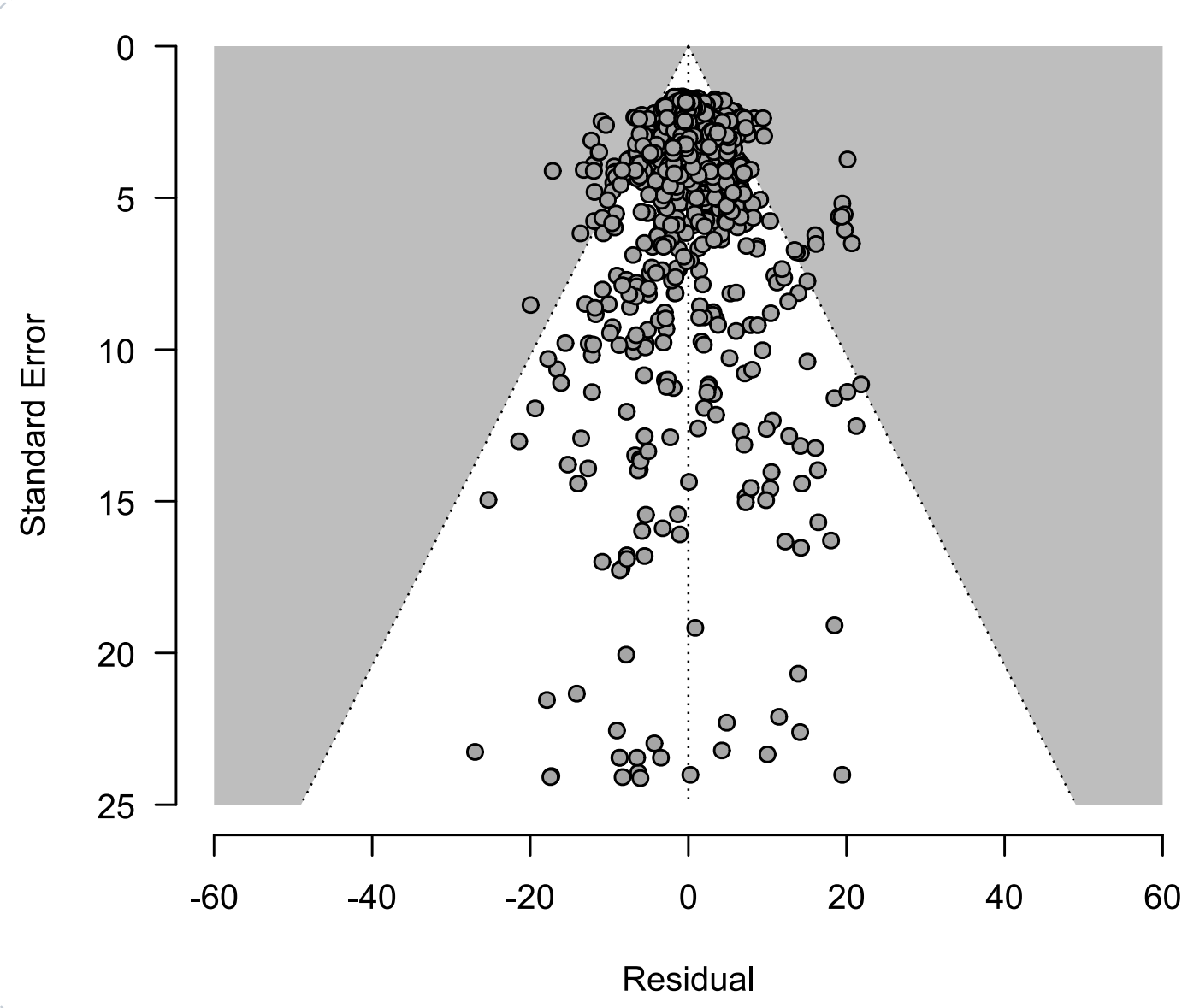
```
mm <- marginal_means(fit)
plot(mm, parameter = "facing_customer", xlim = c(-5, 10),
     lwd = 2, legend_title = "Facing Customer")
```



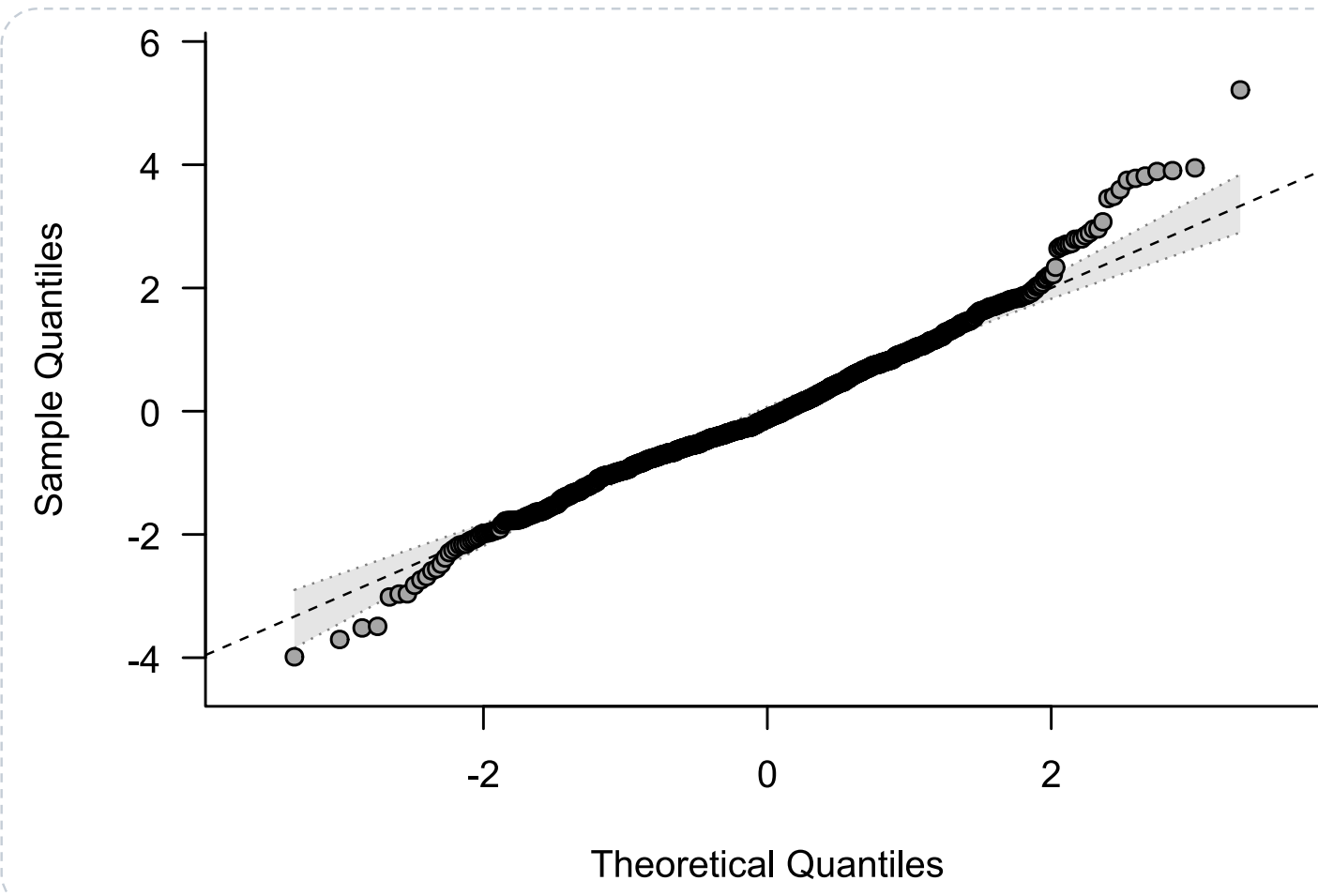
```
> summary_models(fit)

Publication Bias      Prior prob. Post. prob. Inclusion BF
None                  0.500      0.000      0.000
omega[two-sided: .05] 0.042      0.000      0.000
omega[two-sided: .05, .1] 0.042      0.000      0.000
omega[one-sided: .05] 0.042      0.000      0.000
omega[one-sided: .025, .05] 0.042      0.000      0.000
omega[one-sided: .05, .5] 0.042      0.357      12.755
omega[one-sided: .025, .05, .5] 0.042      0.643      41.475
PET                   0.125      0.000      0.000
PEESE                 0.125      0.000      0.000
```

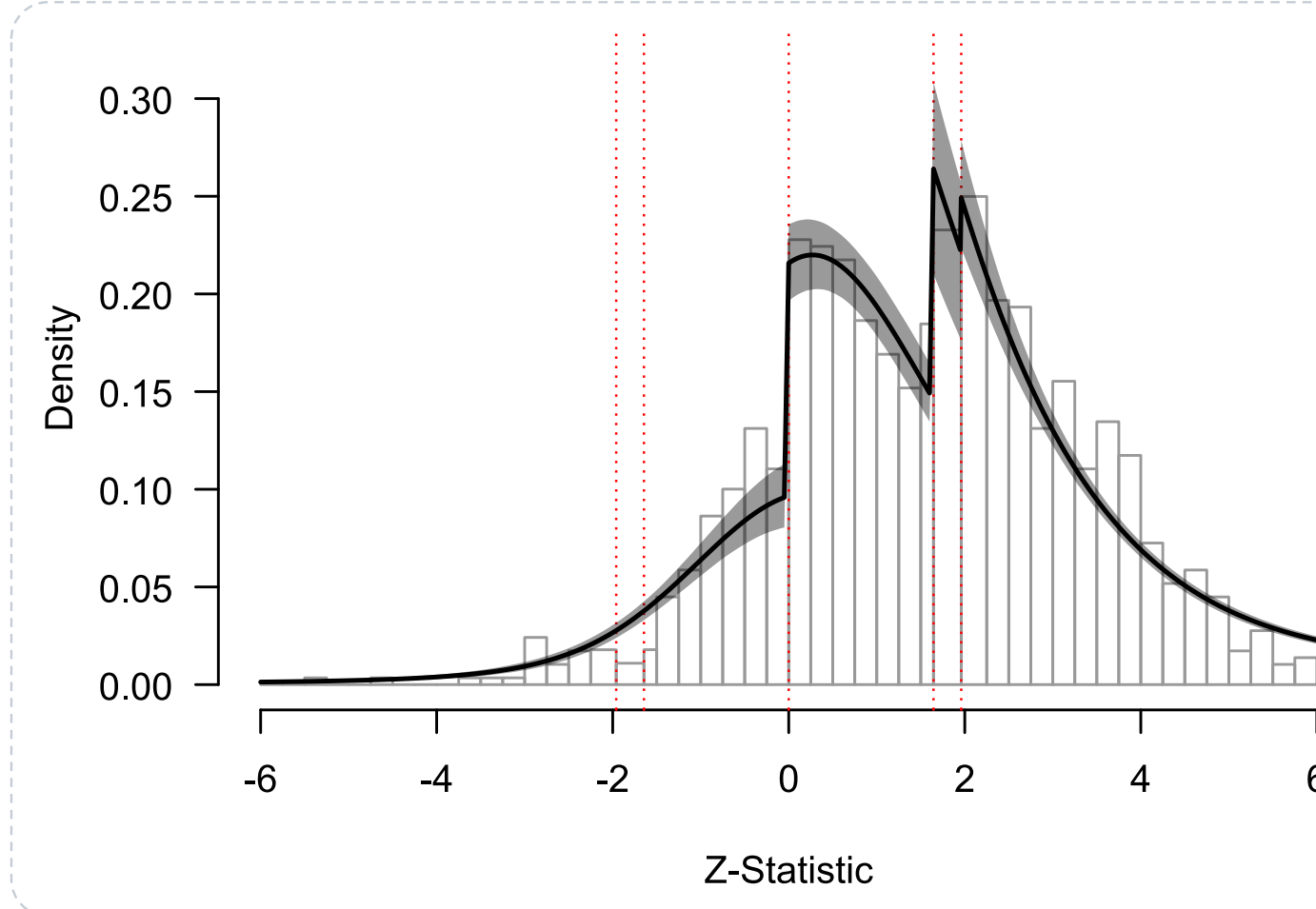
```
fit <- add_loo(fit)
funnel(fit)
```



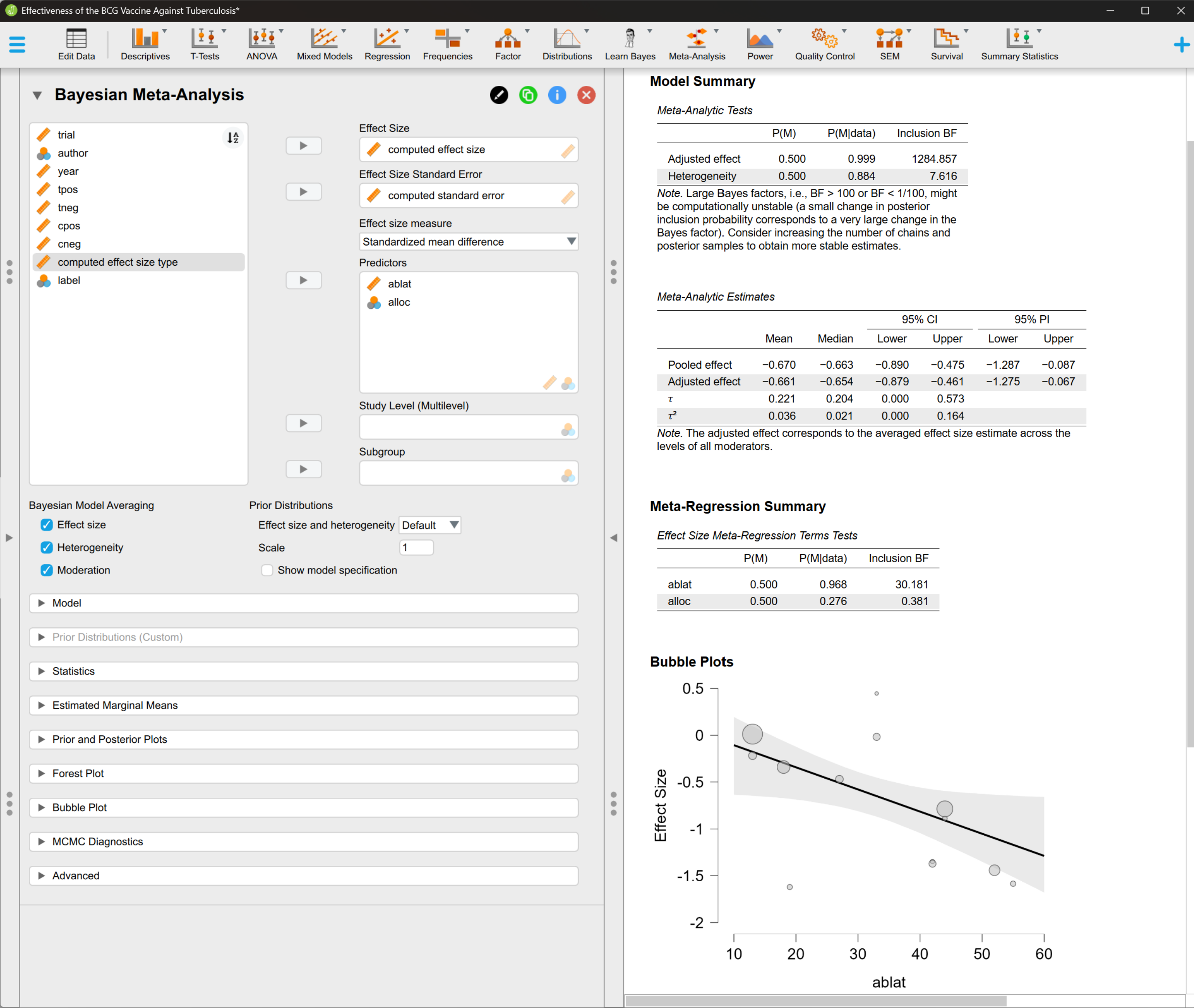
```
qqnorm(fit)
```



```
zplot(fit, lwd = 2, by.hist = 0.25)
```



JASP (https://jasp-stats.org/)



Feature roadmap

RoBMA 4.1 (July 2026)

Multivariate and multilevel meta-analysis
`brma.mv()` · `BMA.mv()` · `RoBMA.mv()`
Hypothesis tests based on posterior distribution
`hypothesis()`
Posterior density estimation
IWMDE, qCMDE

Links

<https://cran.r-project.org/web/packages/RoBMA/>
<https://fbartos.github.io/RoBMA/>

RoBMA 4.2 (September 2026)

Missing data imputation

RoBMA 4.3 (November 2026)

Network meta-analysis

RoBMA 4.4 (2027)

Suggestions?

References

- Maier, M., Bartoš, F., & Wagenmakers, E. J. (2023). Robust Bayesian meta-analysis: Addressing publication bias with model-averaging. *Psychological Methods*, 28(1), 107.
- Bartoš, F., Maier, M., Wagenmakers, E. J., Doucouliagos, H., & Stanley, T. D. (2023). Robust Bayesian meta-analysis: Model-averaging across complementary publication bias adjustment methods. *Research Synthesis Methods*, 14(1), 99-116.
- Bartoš, F., Maier, M., Stanley, T. D., & Wagenmakers, E. J. (2025). Robust Bayesian meta-regression: Model-averaged moderation analysis in the presence of publication bias. *Psychological Methods*.
- Bartoš, F., Maier, M., & Wagenmakers, E. J. (2026). Robust Bayesian multilevel meta-analysis: Adjusting for publication bias in the presence of dependent effect sizes. *Behavior Research Methods*, 58(6), 165.